

COMP438/538 Reinforcement Learning

Introduction

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Brainstorm

- What is intelligence?

“Perceive, infer and retain information and apply it towards goals”

Learning is mostly here.

Planning and Reinforcement Learning (loosely)

- What is artificial intelligence?

“The above but for artificial agents”

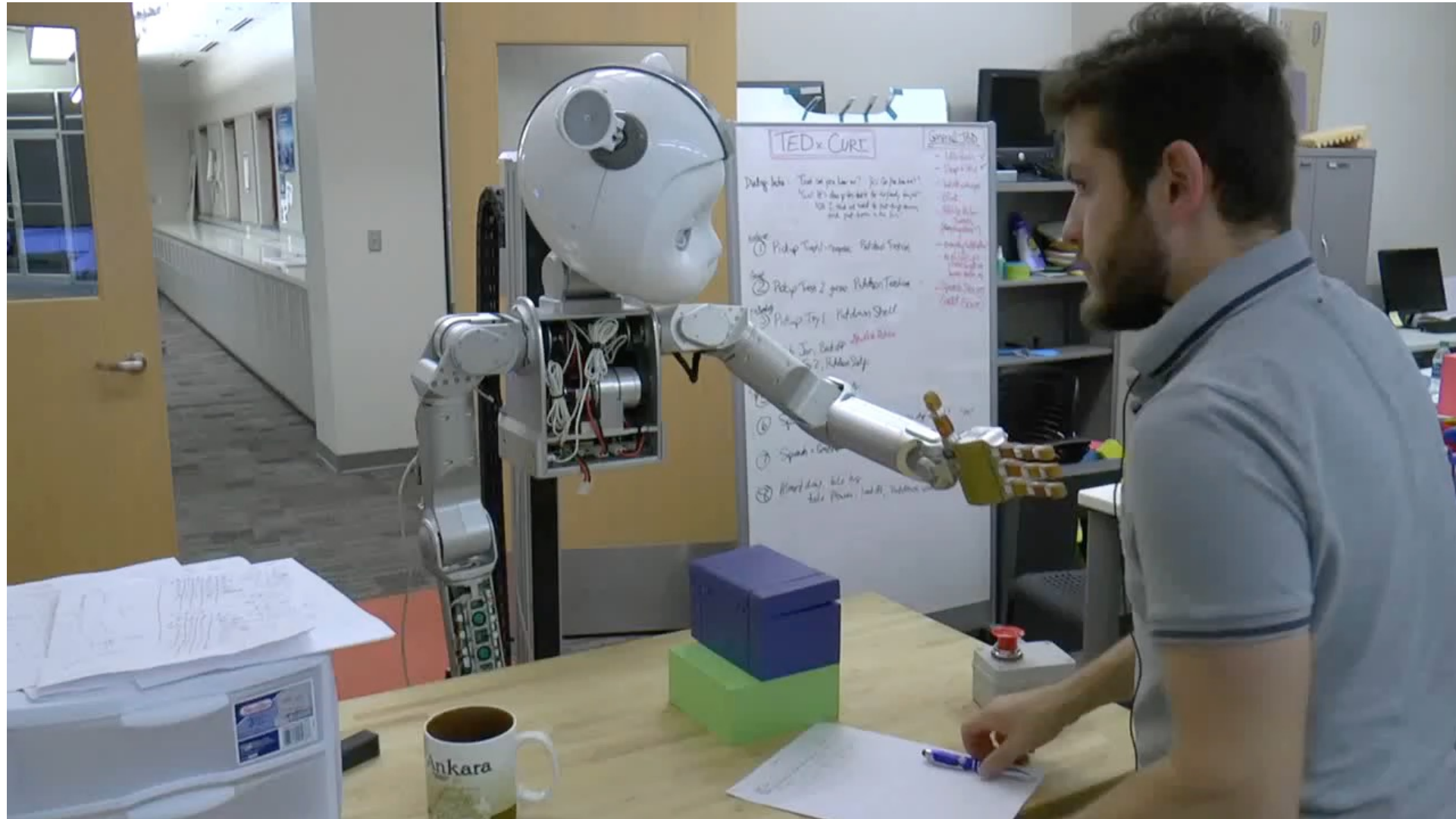
A better alternative: “The science of making **agents** that **act rationally**”

These descriptions lack several fundamental points, but the most important one is computational tractability and efficiency

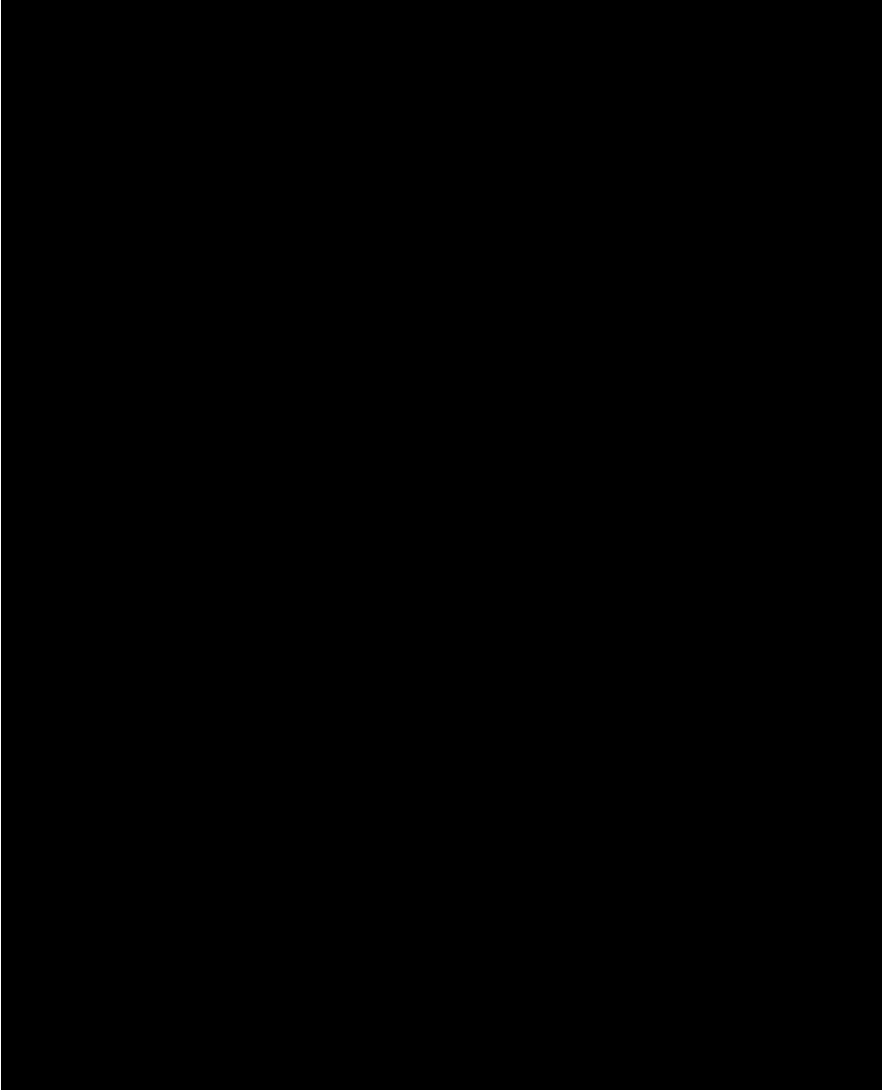
Decision Making

- What “**action**” should I take to maximize my “**expected**” outcome given the current “**situation**”?
- Any ideas...
- In real life:
 - Need to make sequential decisions
 - Outcomes of actions are uncertain
 - There is limited information
 - Rewards can also be uncertain (at least unknown before getting them)
- Reinforcement Learning: “Learning to make decisions by interacting with the environment”

Robots!



Famous Atari Example



In all of these examples, the agents were only given a reward function and time to interact with their environment!

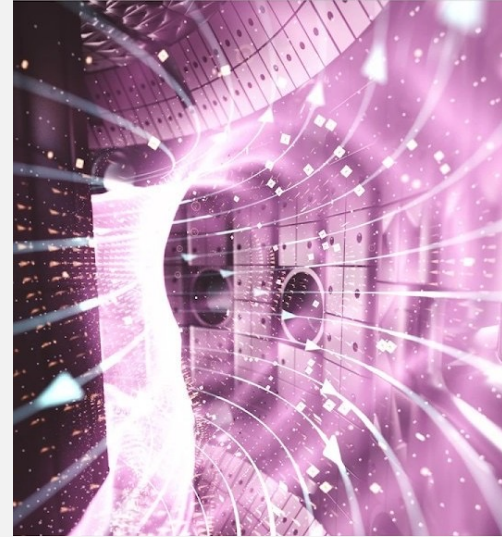
Reinforcement Learning Examples: Why RL?



Go — AlphaGo, 2016

Improving play through self-play

Image: Google DeepMind



Fusion — DeepMind and EPFL, 2022

Learning to control plasma shape

Illustration: Google DeepMind



Robot movement — ANYmal, 2019

Learning movement in simulation, then using it on a real robot

Image: Hwangbo et al., Science Robotics, 2019

Which response is better?

Response A

✓ preferred

Response B

Language — ChatGPT, 2022

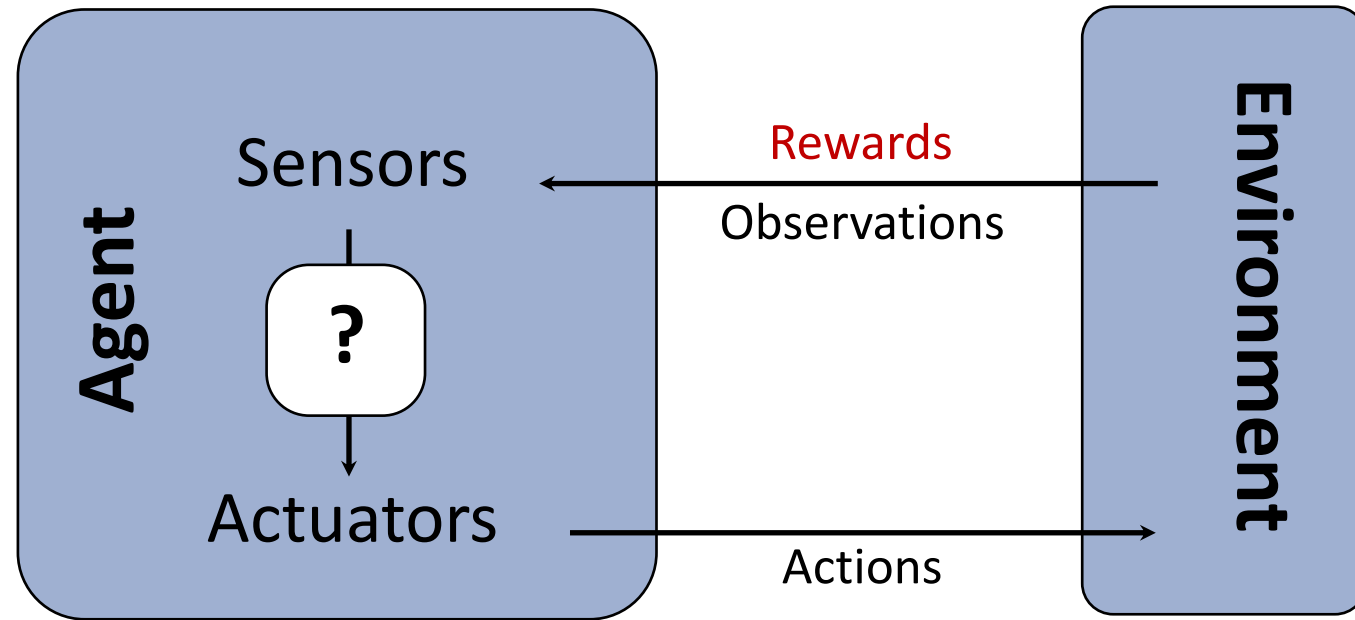
Improving responses with feedback learned from human comparisons

How does RL differ from other learning methods?

Learning setting	Typical learning signal
Supervised learning	Examples of the desired output
Unsupervised learning	Structure in the data, without target labels
Reinforcement learning	Rewards from the consequences of actions

In RL, actions affect later experience. A reward need not tell us which action was best.

Agent and environment



The agent chooses an action. (Based on what?)

The environment responds with an observation and a reward.

A policy describes how the agent chooses actions.

Concrete Example: Recycling Robot

- Mobile robot in an office
- Its task is to collect empty cans
- Solve it with RL
 - “robot”
 - “office”
 - “interaction”
 - “decisions”
 - “learn”

Reward and return

- **Reward:** Feedback for each step (i.e. each time an action is taken).
- **Return:** Added up "future" rewards (later ones may count less).
 - Another interpretation: the rewards still to come
 - Can be defined for each time step!
- Recharging earns nothing immediately, but changes what can happen next. (0 reward yet it is important for success)

Learning from experience

- If the robot is not given the consequences of its actions, how could it learn them?
- On a trial, it observes what happened for the action it chose, not directly what the alternatives would have produced.
- **Exploration:** trying actions to learn more.
- **Exploitation:** using what it has learned to obtain reward.

Observation and state

- **Observation:** what the agent receives now (e.g. through its sensors)
- **State:** a description used to predict consequences and choose actions.
 - A useful state must retain the information that matters for what happens next and ideally not include irrelevant information, sort of an “abstraction”
 - Built from observations
- The battery gauge is hidden. You have just collected two cans.
- **Is the battery high or low?**
- **What earlier events could help you decide what to do next?**

Formulating a decision problem

- For the recycling robot:
 - What decisions does it make, and what feedback does it receive?
 - Why might the latest observation be insufficient?
 - Why doesn't one good outcome settle which behaviour to use?

Concrete Example: Recycling Robot

- Mobile robot in an office
- Its task is to collect empty cans

- Simplified Formulation:

- State/Observations
- How the “world works”
- Actions
- Rewards

- Example *transitions*:

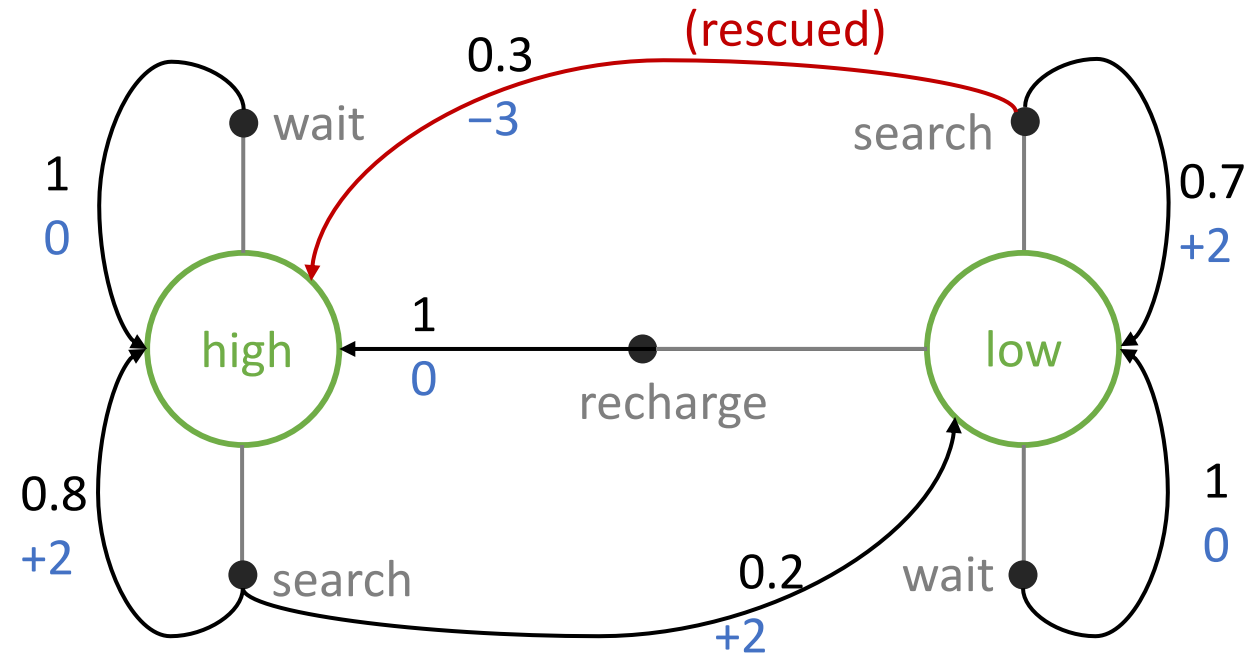
- Always search:

high, search, 2 -> high, search, 2 -> low, search, -3 -> high, search, 2 -> high, ...

- Search when high, recharge when low:

high, search, 2 -> high, search, 2 -> low, recharge, 0 -> high, search, 2 -> high, ...

- Returns?
- Side Question: Which “strategy” is better?



Adapted from Sutton & Barto, Example 3.3. Probabilities and search/wait rewards are illustrative.

Summary: Recycling robot, simplified formulation

Component	Choice in this example
Decisions	Search, wait, recharge
State (same as observation for this example)	High or low battery charge
Model	Search may lower the charge: from high it stays high with 0.8 (else low); from low it stays low with 0.7 (else the battery runs out and the robot is rescued to high). Wait keeps the charge; recharge returns it to high.
Reward	Successful search: 2; wait: 0; recharge: 0; depletion/rescue: -3
Objective	Collect cans over time while accounting for rescue costs
Horizon	Continuing

Why is RL different?

- Learns from **evaluative** feedback (rewards: how good was what I did?), not only **instructive** feedback (labels: what should I have done?)
 - and can also use a teacher when available, e.g. human game records (AlphaGo) or human comparisons (ChatGPT)
- Decisions are made sequentially (agent paradigm) – not i.i.d.!
- Agent's actions affect the subsequent data it receives
- Feedback is usually delayed (action now may have a huge impact in the future)
- **Next:** Define the model, the return objective and the policy.